

Spacetime Dynamics of Altermagnetic Magnons

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Distinguishing altermagnetism from conventional ferromagnetism and antiferromagnetism typically relies on momentum-space probes. Here, we show that the real-space spreading of a localized spin excitation provides a distinctive dynamical fingerprint of altermagnetic order. Using linear spin-wave theory on a two-dimensional checkerboard lattice with nearest-neighbor and next-nearest-neighbor couplings J_1 and J_2 , we demonstrate that finite J_2 produces direction-dependent magnon group velocities and splits the two magnon branches. The resulting propagation remains closer to that of an antiferromagnet than a ferromagnet, retaining an approximately circular outer wavefront, while the direction-dependent magnon velocities produce a pronounced cross-like spatial structure inside this front. We further show that changing the sign of J_2 to enter the unfrustrated regime ($J_2 < 0$) increases the characteristic propagation velocities and interchanges the diagonal directions of enhanced propagation, corresponding to a $\pi/2$ rotation of the anisotropic spatial pattern. These results establish spacetime dynamics as a complementary probe of altermagnetic magnons, providing a route to identifying unconventional magnetic order through real-space propagation patterns in solid-state and synthetic quantum systems.

I. INTRODUCTION

For decades, the study of magnetic materials was defined by a neat dichotomy between ferromagnetism, with its net macroscopic magnetization, and antiferromagnetism, characterized by vanishing net moments and spin-degenerate bands. The recent recognition of altermagnetism has drawn intense interest by establishing a third class of collinear magnetic order—one that combines zero net magnetization with robust spin-splitting [1, 2].

The hallmark signature of altermagnetism is its non-relativistic electronic spin splitting, which can be directly observed in reciprocal space using angle-resolved photoemission spectroscopy (ARPES) [3, 4]. However, for correlated magnets—where conventional ARPES signatures of altermagnetism are fundamentally more challenging to access [5]—and more generally from the perspective that ordered phases are characterized by their collective excitations, a natural question is how collective spin excitations in altermagnets differ from those in ferromagnets and antiferromagnets. Indeed, altermagnetic magnons inherit the underlying crystalline symmetries, acquiring distinct chiral character and momentum-dependent band splittings, as recently demonstrated experimentally in candidate materials such as α -MnTe and CrSb using inelastic neutron scattering (INS) and resonant inelastic x-ray scattering (RIXS) [6, 7]. Yet, resolving these altermagnetic magnon splittings with INS can be experi-

mentally challenging, particularly since the energy and momentum separations of the split chiral magnons are small. In fact, in typical altermagnets they are orders of magnitude smaller than the electronic spin splittings in ARPES.

While such momentum-dependent features naturally make reciprocal-space techniques the standard probe, these experimental challenges motivate complementary approaches. More broadly, the study of quantum many-body systems has increasingly moved toward observing and characterizing dynamics directly in the time domain. Pump-probe and ultrafast spectroscopies have made it possible to follow the evolution of electronic and magnetic excitations following a controlled perturbation [8], while quantum gas microscopes provide a complementary route to resolving and tracking individual spins directly in real space [9, 10]. In parallel, recent theoretical, numerical, and experimental advances have demonstrated that real-space dynamics can reveal nontrivial quantum transport and correlation phenomena, while becoming increasingly accessible through modern computational and quantum-simulation techniques [11–20]. In conventional magnetic systems, such dynamics can produce characteristic real-space propagation patterns, ranging from quantum-walk behavior to “stone-in-a-pond” waves [21], highlighting how real-space dynamics can reveal structure that is not readily apparent in momentum-frequency space.

Importantly, the transition from momentum-frequency to real-space and real-time information does not necessarily require a time-resolved experiment. The dynamical structure factor $S(\mathbf{q}, \omega)$ and its real-space, real-time counterpart are related by a Fourier transform, so con-

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ventional spectroscopic data can in principle be reorganized into a spacetime correlation function $S(\mathbf{r}, t)$. This change of representation can be more than cosmetic: structures that appear as broad continua or overlapping features in $\mathbf{q}$ - ω can acquire a much more organized spatial and temporal form after the transformation. Recent work on the triangular antiferromagnet KYbSe₂ provides a striking example [22]. By Fourier transforming INS data to obtain $S(\mathbf{r}, t)$, the authors observed a nonlinear light cone of spin correlations with dynamics intermediate between ballistic and diffusive spreading. The resulting behavior was not reproduced by several conventional theoretical descriptions and was discussed in connection with exotic collective dynamics in the vicinity of a proximate quantum-spin-liquid regime [12].

This example further confirms a broader principle: the real-space, real-time representation of a magnetic spectrum can expose dynamical organization that is difficult to recognize in momentum-frequency space alone. Therefore, here we apply this perspective to altermagnetism by investigating the spacetime dynamics of localized spin excitations on a two-dimensional checkerboard Heisenberg lattice, where altermagnetic order arises from competing nearest-neighbor (J_1) and next-nearest-neighbor (J_2) exchange couplings [23–25], see Fig. 1(a). A finite J_2 lifts the magnon degeneracy away from symmetry-protected nodal lines and produces momentum-dependent differences in the group velocities of the split branches. By tracking the real-space spreading of an initial single spin flip, we show that while the overall propagation boundary retains the circular footprint of the parent antiferromagnet, the directional velocity mismatch generates a prominent cross-shaped internal interference pattern. Furthermore, we demonstrate that these dynamical signatures depend critically on the sign and magnitude of J_2 , driving a rotation of high-velocity features and changing the characteristic propagation velocities.

The remainder of the paper is organized as follows. In Sec. II, we introduce the 2D spin-1/2 Heisenberg model on the checkerboard lattice, incorporating both nearest-neighbor (J_1) and next-nearest-neighbor (J_2) interactions. Using the Holstein-Primakoff transformation and linear spin-wave theory, we derive the altermagnetic magnon dispersion relations and establish the framework for calculating the time-dependent density profile of a single-spin-flip excitation. In Sec. III, we present our central findings, demonstrating how the distinct spacetime propagation of localized spin excitations provides a dynamical probe of the underlying altermagnetic magnon structure. Specifically, we show that the propagation retains the overall character of the parent antiferromagnet, while the altermagnetic magnon splitting produces a direction- and sublattice-dependent spatial structure whose orientation and propagation velocity depend on the sign and magnitude of J_2 . Finally, in Sec. IV, we summarize our results and discuss their broader implications. The paper is supplemented by two short appendices.

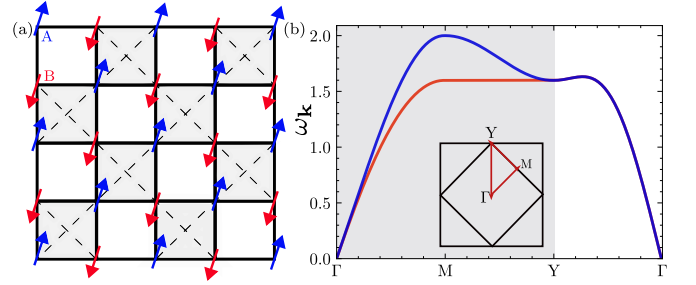


FIG. 1. (a) Schematic representation of the Heisenberg model on a 2D checkerboard lattice featuring collinear Néel order. The A and B sublattices are connected by a composite symmetry operation consisting of a mirror reflection and a spin inversion. (b) Magnon dispersion relations $\omega_{\mathbf{k}}$ for the next-nearest-neighbor Heisenberg exchange $J_2 = 0.2$ (with the nearest neighbor exchange $J_1 = 1$). The introduction of next-nearest-neighbor coupling J_2 unequivocally lifts the band degeneracy across the Brillouin zone, except along the high-symmetry nodal axes ($k_x = 0, k_y = 0$) where the degeneracy is protected by the lattice symmetry.

II. MODEL

We consider a two-dimensional Heisenberg model with $S = 1/2$ spin degrees of freedom:

$$\mathcal{H} = J_1 \sum_{\langle \mathbf{i}, \mathbf{j} \rangle} \mathbf{S}_{\mathbf{i}} \cdot \mathbf{S}_{\mathbf{j}} + J_2 \sum_{\langle\langle \mathbf{i}, \mathbf{j} \rangle\rangle} \mathbf{S}_{\mathbf{i}} \cdot \mathbf{S}_{\mathbf{j}}. \quad (1)$$

Here, J_1 and J_2 denote the nearest- and next-nearest-neighbor interactions, respectively, on the checkerboard lattice; see Fig. 1(a).

Since the main goal of this work is to investigate the altermagnetically ordered state, we restrict our analysis to relatively small values of $|J_2|/|J_1|$. In this regime, the ground state is magnetically ordered [26, 27] and for nonzero J_2 realizes a canonical altermagnet [23–25]. The existence of an ordered ground state allows the low-lying excited states of interest to be described in terms of non-interacting magnons within linear spin-wave theory [28].

We therefore introduce Holstein-Primakoff bosons [28, 29], denoted by a and b for the A and B sublattices, respectively:

$$\begin{aligned} S_{\mathbf{i},A}^z &= S - a_{\mathbf{i}}^\dagger a_{\mathbf{i}}, & S_{\mathbf{i},A}^+ &= \sqrt{2S - a_{\mathbf{i}}^\dagger a_{\mathbf{i}}} a_{\mathbf{i}} = (S_{\mathbf{i},A}^-)^\dagger, \\ S_{\mathbf{i},B}^z &= b_{\mathbf{i}}^\dagger b_{\mathbf{i}} - S, & S_{\mathbf{i},B}^+ &= b_{\mathbf{i}}^\dagger \sqrt{2S - b_{\mathbf{i}}^\dagger b_{\mathbf{i}}} = (S_{\mathbf{i},B}^-)^\dagger. \end{aligned} \quad (2)$$

We then expand the Holstein-Primakoff transformation in powers of $1/\sqrt{2S}$ and retain terms in the Hamiltonian up to quadratic order in the bosonic operators, corresponding to the linear spin-wave approximation (see Appendix A).

Then applying a spatial Fourier transform over the N_c unit cells (with the total number of sites $N_s = 2N_c$), $a_{\mathbf{j}} = (N_c)^{-1/2} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{j}} a_{\mathbf{k}}$ (where $\mathbf{k}$ is the lattice quasi-momentum vector), we can express the Hamiltonian in

momentum space as

$$H = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger H_{\mathbf{k}} \Psi_{\mathbf{k}} + \text{const.}, \quad (3)$$

where $\Psi_{\mathbf{k}}^\dagger = (a_{\mathbf{k}}^\dagger \ b_{-\mathbf{k}})$. The bosonic Bogoliubov-deGennes Hamiltonian matrix $H_{\mathbf{k}}$ mixes the $\pm\mathbf{k}$ modes and takes the form:

$$H_{\mathbf{k}} = \begin{pmatrix} A_{\mathbf{k}} & C_{\mathbf{k}} \\ C_{\mathbf{k}}^* & B_{\mathbf{k}} \end{pmatrix} \quad (4)$$

Where the coefficients $A_{\mathbf{k}} = 4J_1S(1 - 2\eta\Gamma_{\mathbf{k}1})$, $B_{\mathbf{k}} = 4J_1S(1 - 2\eta\Gamma_{\mathbf{k}2})$, and $C_{\mathbf{k}} = 4J_1S\gamma_{\mathbf{k}}$ depend on the lattice geometry, and we define the dimensionless ratio $\eta = J_2/4J_1$, alongside the structure factors $\Gamma_{\mathbf{k}1(2)} = 1 - \cos(k_x \pm k_y)$ and $\gamma_{\mathbf{k}} = [\cos(k_x) + \cos(k_y)]/2$.

This momentum-space Hamiltonian is diagonalized via a Bogoliubov transformation, yielding the diagonal form [23, 25, 30]:

$$H = \sum_{\mathbf{k},\sigma} \omega_{\mathbf{k}\sigma} \left(\alpha_{\mathbf{k}\sigma}^\dagger \alpha_{\mathbf{k}\sigma} + \frac{1}{2} \right) + \text{const.} \quad (5)$$

Here, $\alpha_{\mathbf{k}\sigma}^\dagger$ is the Bogoliubov magnon creation operator. The magnon bands are given by

$$\omega_{\mathbf{k}\sigma} = \sqrt{\left(\frac{A_{\mathbf{k}} + B_{\mathbf{k}}}{2}\right)^2 - |C_{\mathbf{k}}|^2} - \sigma \left(\frac{A_{\mathbf{k}} - B_{\mathbf{k}}}{2}\right), \quad (6)$$

where $\sigma = \pm 1$. The Bogoliubov coefficients, which will be needed below, are

$$\begin{aligned} u_{\mathbf{k}} &= \cosh \left[-\frac{1}{2} \tanh^{-1} \left(\frac{2|C_{\mathbf{k}}|}{A_{\mathbf{k}} + B_{\mathbf{k}}} \right) \right], \\ v_{\mathbf{k}} &= \sinh \left[-\frac{1}{2} \tanh^{-1} \left(\frac{2|C_{\mathbf{k}}|}{A_{\mathbf{k}} + B_{\mathbf{k}}} \right) \right]. \end{aligned} \quad (7)$$

Further details of the Bogoliubov transformation are provided in Appendix A.

Although the magnon bands $\omega_{\mathbf{k}\sigma}$ are perfectly degenerate in the purely nearest-neighbor limit ($J_2 = 0$), the introduction of a finite $J_2 \neq 0$ lifts this degeneracy. As depicted in Fig. 1b, the band splitting is observed across the Brillouin zone, save for the nodal lines $k_x = 0$ and $k_y = 0$. Along these high-symmetry axes, the degeneracy is protected by the underlying symmetries of both the Hamiltonian H and the ground state—specifically, their invariance under a composite transformation that couples spin inversion with a mirror reflection interchanging sublattices A and B [31]. Furthermore, the branch index σ in the dispersion relation $\omega_{\mathbf{k}\sigma}$ serves as a well-defined quantum number. It indicates the net change in the macroscopic magnetization induced by the excitation of a single magnon, yielding $\Delta S_{\text{tot}}^z = \sigma$ [30].

With the model diagonalized, we now examine the real-time evolution of the system following a single spin-flip excitation at a specific site $\mathbf{r}_o$, applied to the Hamiltonian

ground state, i.e. the Bogoliubov magnon vacuum $|\emptyset\rangle$. We define the time-evolved state as

$$|\phi(\mathbf{r}, t)\rangle = c_{\mathbf{r}} e^{-iHt/\hbar} c_{\mathbf{r}_o}^\dagger |\emptyset\rangle, \quad (8)$$

where the annihilation operator $c_{\mathbf{r}}$ corresponds to either $a_{\mathbf{r}}$ or $b_{\mathbf{r}}$ depending on the parity of site $\mathbf{r}$. This leads to four distinct scenarios based on the sublattice locations of $\mathbf{r}$ and $\mathbf{r}_o$. Explicitly, when $\mathbf{r}_o$ lies in the A sublattice, for $\mathbf{r} \in A$, we have

$$\begin{aligned} |\phi(\mathbf{r}, t)\rangle &= \frac{1}{N_c} \sum_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k},-} t)} u_{\mathbf{k}}^2 |\emptyset\rangle \\ &+ \frac{1}{N_c} \sum_{\mathbf{k}\mathbf{k}'} e^{i(\mathbf{k}' \cdot \mathbf{r} - \mathbf{k} \cdot \mathbf{r}_o - \omega_{\mathbf{k},-} t)} \\ &\times u_{\mathbf{k}} v_{\mathbf{k}'} |\alpha_{\mathbf{k},-}, \alpha_{-\mathbf{k}',+}\rangle \end{aligned} \quad (9)$$

and for $\mathbf{r} \in B$, we have

$$\begin{aligned} |\phi(\mathbf{r}, t)\rangle &= \frac{1}{N_c} \sum_{\mathbf{k}\mathbf{k}'} e^{i(\mathbf{k}' \cdot \mathbf{r} - \mathbf{k} \cdot \mathbf{r}_o - \omega_{\mathbf{k},-} t)} \\ &\times \sqrt{1 + \delta_{\mathbf{k},-\mathbf{k}'} u_{\mathbf{k}} v_{\mathbf{k}'}} |\alpha_{\mathbf{k},-}, \alpha_{-\mathbf{k}',-}\rangle \end{aligned} \quad (10)$$

with similar expressions for $\mathbf{r}_o \in B$ given in Appendix B.

Finally, the main goal of the paper is to track the observable probability densities, i.e. the density profile ρ defined as

$$\rho(\mathbf{r}, t) = \langle \phi(\mathbf{r}, t) | \phi(\mathbf{r}, t) \rangle. \quad (11)$$

Then, we evaluate that for an initial excitation on the A sublattice ($\mathbf{r}_o \in A$) the density profile is given by:

$$\begin{aligned} \rho_A(\mathbf{r}, t) &= \frac{1}{N_c^2} \left| \sum_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k},-} t)} u_{\mathbf{k}}^2 \right|^2 \\ &+ \frac{2}{N_c^2} \left(\sum_{\mathbf{k}} u_{\mathbf{k}}^2 \right) \left(\sum_{\mathbf{k}} v_{\mathbf{k}}^2 \right) \\ &+ \frac{2}{N_c^2} \left(\sum_{\mathbf{k}} u_{\mathbf{k}}^2 v_{\mathbf{k}}^2 \right) \\ &+ \frac{1}{N_c^2} \left| \sum_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k},-} t)} u_{\mathbf{k}} v_{\mathbf{k}} \right|^2. \end{aligned} \quad (12)$$

Similar expressions for $\mathbf{r}_o \in B$, i.e. for $\rho_B(\mathbf{r}, t)$, are given in Appendix B.

III. RESULTS

A. Ferromagnetic and antiferromagnetic limits

We start by revisiting the simplest case in which next-nearest-neighbor interactions are absent ($J_2 = 0$) and the spin exchange is ferromagnetic ($J_1 < 0$), see [21]. The ground state is then fully polarized and acts as a true

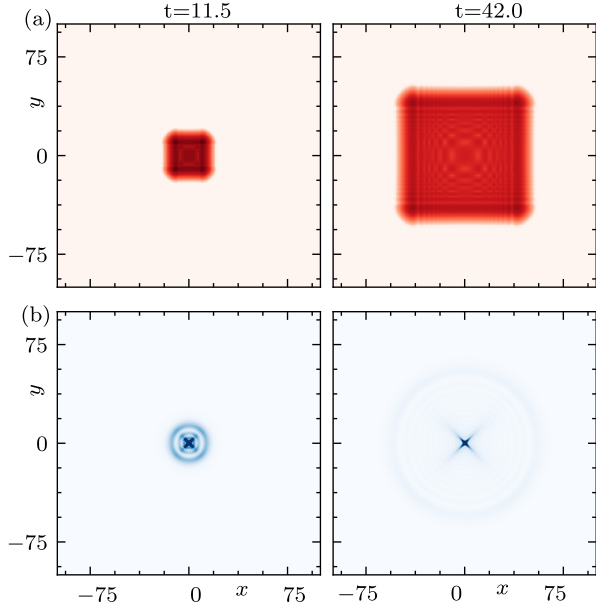


FIG. 2. Comparison of the real-space spreading of a single-site spin-flip excitation in the absence of next-nearest-neighbor interactions ($J_2 = 0$). (a) In the ferromagnetic case ($J_1 < 0$), the largest group velocities occur at finite momenta, making the propagation sensitive to short-wavelength, lattice-scale physics and producing the characteristic square-shaped density profile. (b) In the antiferromagnetic case ($J_1 > 0$), the largest group velocities occur in the long-wavelength regime, where the linear magnon dispersion is approximately rotationally symmetric. The resulting wavefront is therefore nearly circular and much less sensitive to the underlying square-lattice structure.

vacuum for magnons, with no quantum fluctuations. The magnon excitations are characterized by the well-known ferromagnetic dispersion $\omega_{\mathbf{k}} = 4S|J_1|(1 - \gamma_{\mathbf{k}})$, which is quadratic in momentum, $\omega_{\mathbf{k}} \propto k^2$, near the center of the Brillouin zone (the Γ point). Because the highest magnon group velocities originate from regions of the Brillouin zone away from its center, the spatial spreading of a localized spin flip is dominated by large momenta and thus by short-wavelength, lattice-scale physics that strongly reflects the underlying lattice symmetry [21]. Consequently, the dynamics following a single spin flip map onto a continuous-time “quantum walk” of a single particle, producing the characteristic square-shaped spatial density profile shown in Fig. 2(a).

The situation is qualitatively different in the antiferromagnet ($J_1 > 0$ and $J_2 = 0$), whose ordered ground state is dressed by quantum fluctuations. The corresponding magnon dispersion is $\omega_{\mathbf{k}} = 4SJ_1\sqrt{1 - \gamma_{\mathbf{k}}^2}$. Near the Γ point, it becomes linear in momentum, $\omega_{\mathbf{k}} \propto |\mathbf{k}|$, and hence the maximum group velocity is reached in this long-wavelength limit. The spreading of a localized spin flip is therefore dominated by long-wavelength excitations and resembles the propagation of a “stone-in-a-pond” wave [21]. Since the long-wavelength dispersion is approximately rotationally symmetric, the resulting

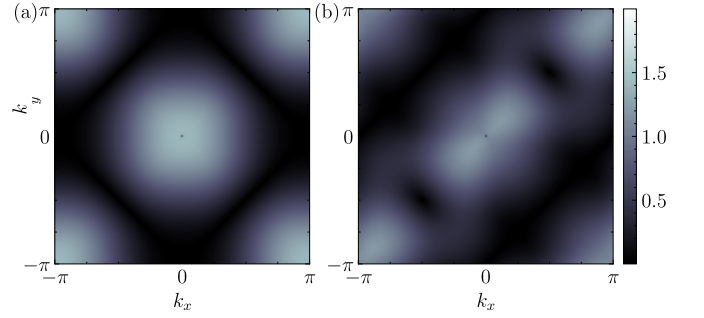


FIG. 3. Magnitude of the magnon group velocity, $|\mathbf{v}_{\sigma}(\mathbf{k})|$, across the Brillouin zone. (a) In the antiferromagnetic limit, $J_2 = 0$, the largest group velocities are distributed nearly isotropically around the Γ point. (b) For finite $J_2 > 0$, this distribution becomes direction dependent, with the largest velocities concentrated along selected diagonal directions. This deformation of the group-velocity distribution reflects the direction dependence of the altermagnetic magnon dispersion discussed in the main text. The parameters are (a) $\theta = 0$ ($J_1 = 1$, $J_2 = 0$) and (b) $\theta = 0.3$ ($J_1 \simeq 0.95$, $J_2 \simeq 0.29$), with $\sigma = -1$.

wavefront is nearly circular and much less sensitive to the underlying square-lattice structure [see Fig. 2(b)].

B. Altermagnetic magnon propagation

Having established the ferromagnetic and antiferromagnetic limits, we now consider the altermagnetic regime obtained for finite J_2 . We parameterize the couplings as $J_1 = \cos \theta$ and $J_2 = \sin \theta$, keeping their overall scale fixed. We first focus on a representative altermagnetic case obtained at $\theta = 0.3$, corresponding to dominant $J_1 \simeq 0.95$ and somewhat weaker $J_2 \simeq 0.29$.

A finite J_2 modifies the magnon dispersion in two related ways. First, it changes the part of the dispersion common to both magnon branches. In the antiferromagnetic limit $J_2 = 0$, the long-wavelength dispersion around the Γ point is approximately rotationally symmetric (despite the underlying square lattice), depending to leading order only on $|\mathbf{k}|$. For $J_2 \neq 0$, the next-nearest-neighbor interactions introduce an additional dependence on the lattice diagonals through $\cos(k_x + k_y)$ and $\cos(k_x - k_y)$. This deforms the nearly isotropic antiferromagnetic dispersion and makes its slope direction dependent. We refer to this common part of the two magnon branches as the *base dispersion*. Since the magnon group velocity is

$$\mathbf{v}_{\sigma}(\mathbf{k}) = \nabla_{\mathbf{k}} \omega_{\mathbf{k}\sigma}, \quad (13)$$

this deformation directly produces a direction-dependent contribution to the propagation velocity.

Second, the diagonal J_2 couplings enter differently for the two sublattices and lift the degeneracy of the two magnon branches away from the symmetry-protected

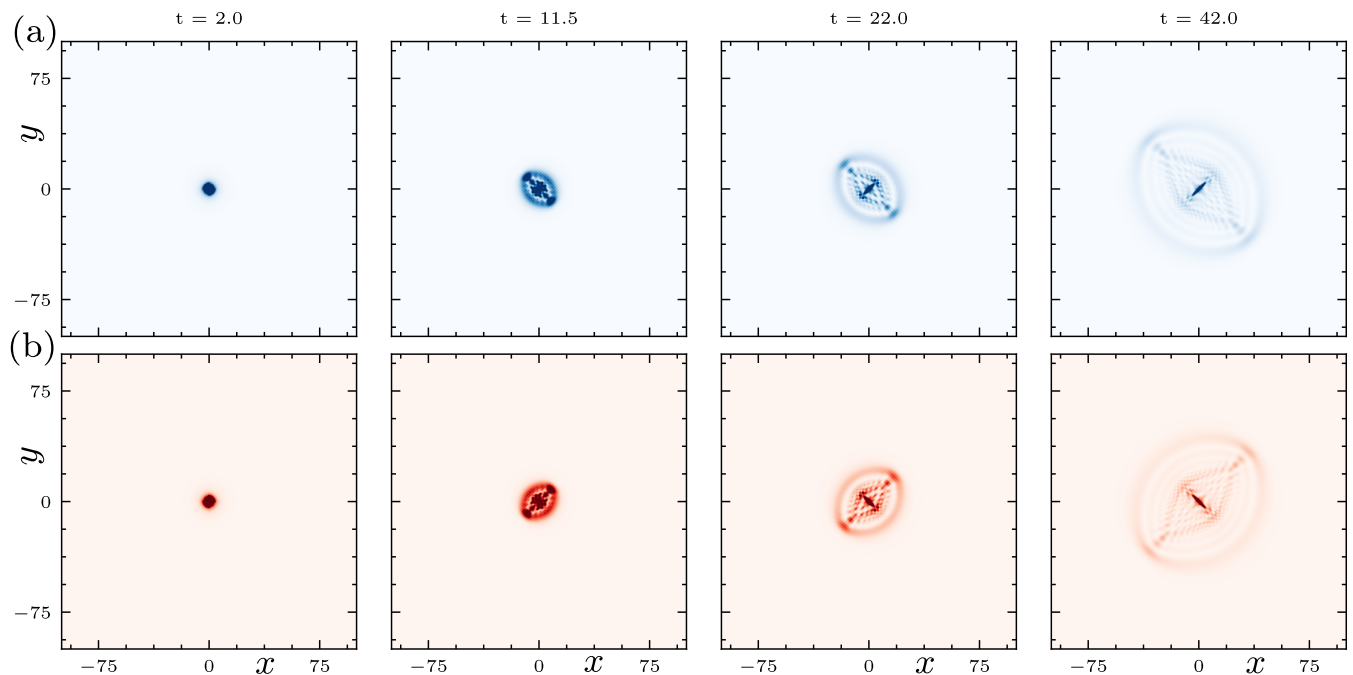


FIG. 4. Time-resolved spatial density profiles following a single-site spin flip in the representative altermagnetic regime, $\theta = 0.3$ ($J_1 \simeq 0.95$, $J_2 \simeq 0.29$), at $t = 2.0, 11.5, 22.0$, and 42.0 . The outer propagation front remains approximately circular, reflecting the dominant antiferromagnetic-like propagation, while the direction-dependent base dispersion and the altermagnetic branch splitting produce a cross-like structure inside the wave packet. Panels (a) and (b) show the $\sigma = +1$ and $\sigma = -1$ magnon branches, respectively. The lattice size is $N = 200$.

nodal lines. This altermagnetic splitting gives an additional, branch-dependent contribution to the group velocity. Thus, the dynamics for finite J_2 reflect two effects: the directional dependence of the base dispersion and the difference between the two split magnon branches.

Figure 3 illustrates these effects explicitly in the calculated group velocities. As discussed in Sec. III A, for $J_2 = 0$ the distribution of the largest group velocities is nearly isotropic around the Γ point. For finite J_2 , this distribution is deformed, with the largest velocities concentrated along selected diagonal directions. In addition, the two magnon branches acquire different velocities away from the nodal lines.

The consequences are directly visible in spacetime magnon dynamics. A spin flip localized on a single lattice site contains all momenta, and its time evolution therefore probes the full magnon dispersion. Figure 4 shows the resulting density profile for $\theta = 0.3$. Since J_1 remains the dominant coupling, the outer propagation front remains approximately circular, as in the parent antiferromagnet. Inside this front, however, the direction-dependent base velocity together with the altermagnetic branch splitting produces the cross-like structure visible in the density profile. The coexistence of a nearly circular outer front with a structured interior is therefore a characteristic real-space consequence of the altermagnetic magnon dispersion.

Overall, the propagation in the altermagnets remains

much closer to that of the parent antiferromagnet than to the ferromagnetic case discussed in Sec. III A. In particular, the outer wavefront remains approximately circular, reflecting the dominant long-wavelength antiferromagnetic propagation. The main effect of altermagnetism is therefore not to replace this behavior, but to superimpose a pronounced direction-dependent structure inside the wave packet. In this limited sense, the internal pattern is reminiscent of the ferromagnetic case, where finite-momentum modes also make the propagation sensitive to the underlying lattice. The crucial difference is that in the altermagnet this lattice-sensitive structure appears within an otherwise antiferromagnetic-like expanding wavefront.

C. Role of frustration

We next examine how the sign of J_2 affects altermagnetic magnon propagation by comparing the frustrated ($J_2 > 0$) and unfrustrated ($J_2 < 0$) regimes. Figure 5 shows two main changes that encountered once J_2 is no longer frustrating J_1 . First, the wave packet spreads considerably farther in the unfrustrated regime, indicating larger characteristic propagation velocities. Second, the directional structure of the wave packet is interchanged between the two lattice diagonals, corresponding to a $\pi/2$ rotation of the anisotropic spatial pattern.

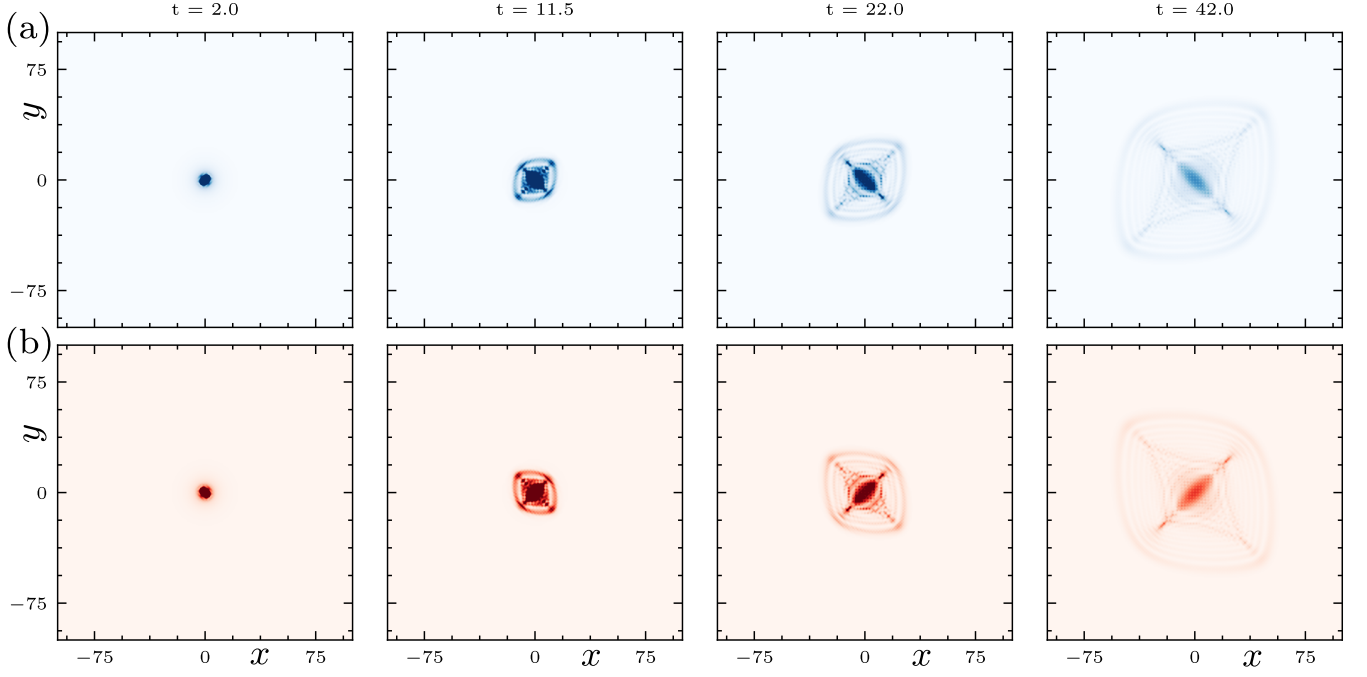


FIG. 5. Time-resolved spatial density profiles following a single-site spin flip in the unfrustrated altermagnetic regime, $\theta = -0.4$ ($J_1 \simeq 0.92$, $J_2 \simeq -0.38$), at $t = 2.0, 11.5, 22.0$, and 42.0 . Compared with the frustrated regime shown in Fig. 4, the wave packet spreads considerably farther and its anisotropic spatial pattern is rotated by $\pi/2$. These changes reflect the larger characteristic magnon group velocities for $J_2 < 0$ and the reversal of the branch-dependent altermagnetic contribution discussed in the main text. Panels (a) and (b) show the $\sigma = +1$ and $\sigma = -1$ magnon branches, respectively. The lattice size is $N = 200$.

The sublattice dependence of the latter effect is resolved more clearly in Fig. 6, which shows the density at $t = 11.5$ along the diagonal $y = x$. For $J_2 > 0$, the wave packet extends farther along this diagonal on sublattice A than on sublattice B, whereas reversing the sign of J_2 interchanges this behavior. Thus, changing the sign of J_2 reverses which sublattice exhibits the enhanced propagation along a given diagonal. Figure 6 also shows that the overall spatial distribution is broader for $J_2 < 0$.

The difference in propagation speed is also nicely seen in the so-called light cones. Figure 7 shows the “continuous” time evolution of the magnon density profile along the principal and diagonal lattice directions. The different slopes of the propagating fronts demonstrate that the broader spatial profiles are associated with different propagation velocities rather than only with a redistribution of spectral weight. In particular, the unfrustrated case displays a substantially wider propagation cone.

Taken together, Figs. 5–7 therefore reveal two effects of changing the sign of J_2 : a change in the characteristic propagation speed and a reversal of the directional, sublattice-dependent structure. The origin of these two effects can be understood directly from the magnon dispersion. Equation (6) can be written as

$$\omega_{\mathbf{k}\sigma} = \omega_{\text{base}}(\mathbf{k}) + \sigma S J_2 (\Gamma_{\mathbf{k}1} - \Gamma_{\mathbf{k}2}), \quad (14)$$

where

$$\omega_{\text{base}}(\mathbf{k}) = \sqrt{[4J_1 S - S J_2 (\Gamma_{\mathbf{k}1} + \Gamma_{\mathbf{k}2})]^2 - |C_{\mathbf{k}}|^2}. \quad (15)$$

The group velocity can therefore be separated into

$$\mathbf{v}_{\sigma}(\mathbf{k}) = \mathbf{v}_{\text{base}}(\mathbf{k}) + \mathbf{v}_{\text{alt},\sigma}(\mathbf{k}), \quad (16)$$

where the two terms account for the overall propagation scale and the branch-dependent directional structure, respectively.

We first consider the base velocity. Near the Γ point,

$$\Gamma_{\mathbf{k}1} + \Gamma_{\mathbf{k}2} \simeq k^2, \quad \gamma_{\mathbf{k}} \simeq 1 - \frac{k^2}{4}, \quad (17)$$

so that

$$\omega_{\text{base}}(\mathbf{k}) \simeq S \sqrt{8J_1(J_1 - J_2)} |\mathbf{k}|. \quad (18)$$

The corresponding long-wavelength group velocity is

$$v_{\text{base}} \simeq S \sqrt{8J_1(J_1 - J_2)}. \quad (19)$$

For $J_1 > 0$, this velocity increases when $J_2 < 0$ and decreases when $J_2 > 0$. Thus, the unfrustrated next-nearest-neighbor interaction enhances the long-wavelength antiferromagnetic-like propagation, whereas frustration suppresses it. Although intuitively such result

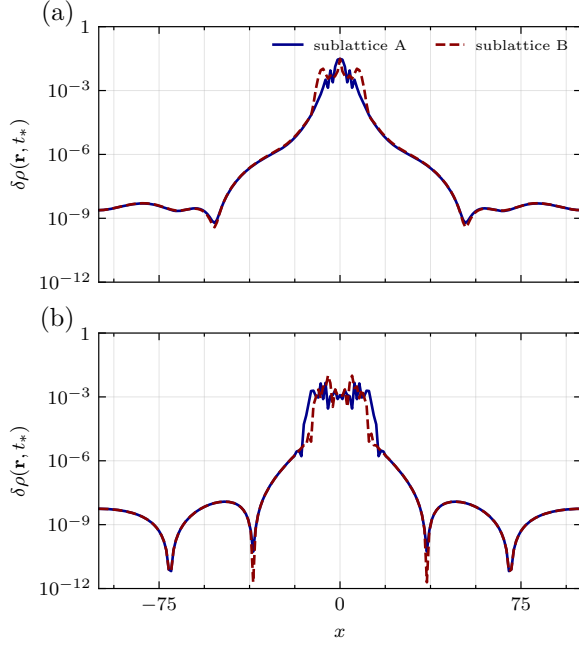


FIG. 6. Sublattice-resolved density profiles at $t = 11.5$ along the diagonal $y = x$ for (a) the frustrated regime, $\theta = 0.3$, and (b) the unfrustrated regime, $\theta = -0.3$. Reversing the sign of J_2 interchanges the enhanced propagation between the A and B sublattices along this diagonal. The spatial distribution is also broader for $J_2 < 0$, reflecting the larger characteristic propagation velocity in the unfrustrated regime.

could have been expected, this provides a direct explanation for the faster spreading observed in the unfrustrated regime.

The directional and sublattice-dependent behavior originates from the second term in Eq. (14). Using

$$\Gamma_{\mathbf{k}1} - \Gamma_{\mathbf{k}2} = 2 \sin k_x \sin k_y, \quad (20)$$

its contribution to the group velocity is

$$\mathbf{v}_{\text{alt},\sigma}(\mathbf{k}) = 2\sigma S J_2 (\cos k_x \sin k_y, \sin k_x \cos k_y). \quad (21)$$

This contribution changes sign under either $J_2 \rightarrow -J_2$ or $\sigma \rightarrow -\sigma$. Reversing J_2 therefore interchanges the branch-dependent velocities. Since the sublattice-resolved dynamics are tied to opposite magnon branches, this accounts for the interchange between A and B observed in Fig. 6.

Finally, the same term explains the rotation of the directional pattern. Under a $\pi/2$ rotation, $(k_x, k_y) \rightarrow (-k_y, k_x)$,

$$\sin k_x \sin k_y \rightarrow -\sin k_x \sin k_y. \quad (22)$$

A $\pi/2$ rotation therefore reverses the branch-dependent part of the dispersion in the same way as $J_2 \rightarrow -J_2$. This explains why reversing the sign of J_2 interchanges the two diagonal directions of enhanced propagation seen in Fig. 5.

IV. CONCLUSION

A. Summary

In summary, we have theoretically characterized the spacetime dynamics of localized spin excitations in a two-dimensional checkerboard altermagnet and compared their propagation with the conventional ferromagnetic and antiferromagnetic limits. Using the Holstein–Primakoff transformation and linear spin-wave theory, we showed that finite next-nearest-neighbor interactions (J_2) lift the magnon branch degeneracy away from symmetry-protected nodal lines and produce momentum- and branch-dependent group velocities.

These features manifest directly in the real-time spreading of a localized spin flip. In the altermagnetic regime with $J_2 > 0$, the outer propagation front remains approximately circular, retaining the dominant long-wavelength character of the parent antiferromagnet. Inside this front, however, the direction-dependent base dispersion and the splitting of the two magnon branches produce a pronounced cross-like spatial structure. The coexistence of an antiferromagnetic-like outer wavefront with a direction-dependent internal pattern therefore provides a characteristic real-space signature of the altermagnetic magnon dispersion.

We further showed that the propagation depends strongly on the sign of J_2 . In the unfrustrated regime ($J_2 < 0$), the characteristic propagation velocity is enhanced relative to the frustrated case. This follows directly from the long-wavelength base velocity, $v_{\text{base}} \propto \sqrt{J_1(J_1 - J_2)}$, which increases for $J_2 < 0$ and decreases for $J_2 > 0$. At the same time, the branch-dependent contribution to the group velocity is proportional to σJ_2 and therefore reverses upon changing the sign of J_2 . This accounts for the interchange of the sublattice-dependent propagation and the $\pi/2$ rotation of the directional spatial pattern. Together, these results show how both the magnon propagation speed and its directional structure encode the underlying altermagnetic interactions.

B. Experimental outlook

The spacetime patterns identified here are, in principle, accessible to bulk spectroscopies, with the two-sublattice structure providing an additional experimental handle beyond the total dynamical structure factor. In a unit cell containing two sublattices, the INS cross section contains interference terms between basis sites, with their relative phase determined by the momentum transfer $\mathbf{q}$. Consequently, measurements at different reciprocal-lattice positions probe distinct combinations of intra- and inter-sublattice dynamical correlations. A sufficiently complete single-crystal dataset of the $(\mathbf{q}, \omega)$ -dependent response can therefore be used to disentangle these basis contributions and, after Fourier transformation, reconstruct sublattice-resolved real-space correla-

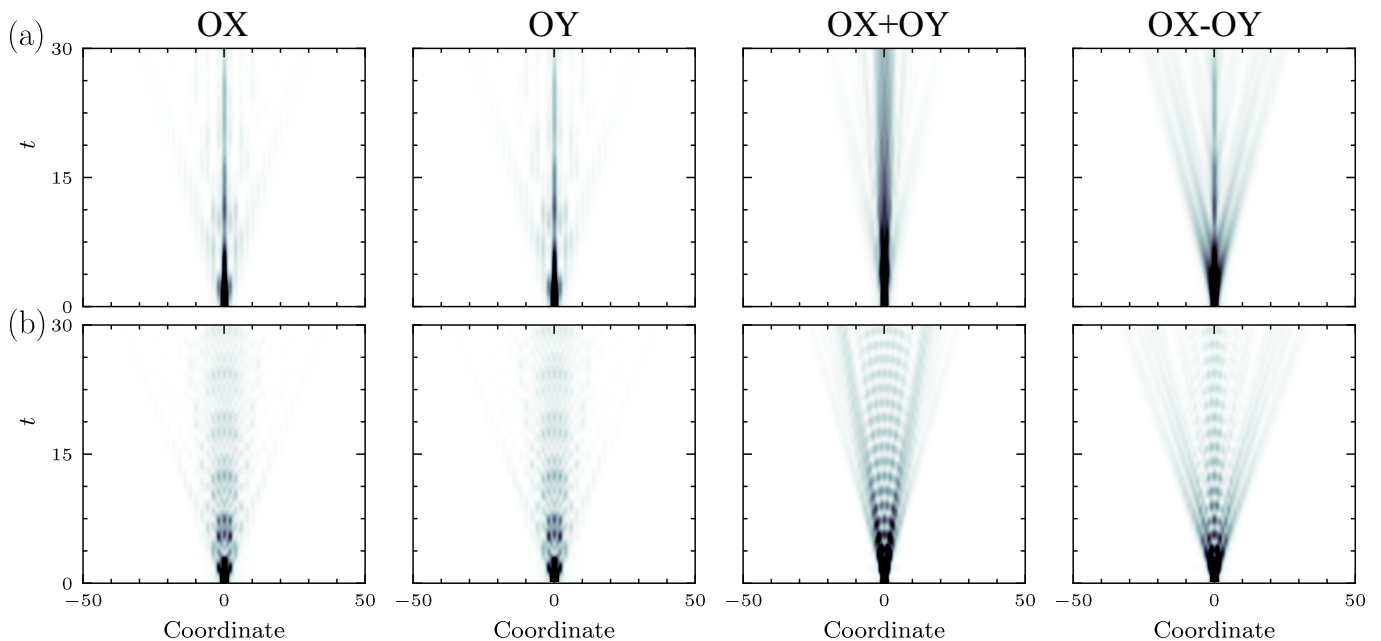


FIG. 7. Spatiotemporal density profiles following a single-site spin flip along the principal (OX, OY) and diagonal (OX+OY, OX-OY) lattice directions. (a) Frustrated regime, $\theta = 0.3$ ($J_1 \simeq 0.96$, $J_2 \simeq 0.30$), for the $\sigma = +1$ magnon branch. (b) Unfrustrated regime, $\theta = -0.4$ ($J_1 \simeq 0.92$, $J_2 \simeq -0.39$), for the $\sigma = -1$ magnon branch. The broader propagation cones in (b) illustrate the larger characteristic magnon velocities in the unfrustrated regime. The lattice size is $N = 200$.

tion functions. Polarization analysis can provide further separation of spin and chiral components, while the momentum dependence of the magnetic structure factor carries the essential information about the basis-site interference. Although such a reconstruction requires broader momentum coverage than a conventional Fourier transform of the total structure factor $S(\mathbf{q}, \omega)$, it establishes that the sublattice-resolved spacetime dynamics considered here are, in principle, experimentally accessible in bulk solids rather than being restricted to an idealized theoretical construction.

RIXS provides a complementary route in which the sublattice sensitivity can be built directly into the scattering matrix element. Unlike neutron scattering, magnetic RIXS can have amplitudes that depend sensitively on the local orbital environment and photon polarization, allowing different components of the magnetic response to be selectively enhanced through interference between contributions from inequivalent sites [32]. This mechanism is particularly relevant to altermagnetic systems, where the magnetic sublattices are related by nontrivial crystal symmetries rather than by a simple translational operation. Recent RIXS studies of altermagnetic magnons have demonstrated sensitivity to their sublattice and chiral character [7], while sublattice-specific circular dichroism has more recently been proposed and observed in the zero-moment altermagnet $\text{Fe}_2\text{Mo}_3\text{O}_8$ [33]. These developments suggest that, for an altermagnetic lattice such as the checkerboard model considered here, an appropriately chosen RIXS geometry and polarization could assign different weights to the two magnetic sublattices

and thereby enhance the distinct magnon branches. RIXS could thus provide an experimentally complementary view of the same momentum-dependent sublattice structure that, in our calculation, produces the characteristic real-space interference pattern.

A particularly direct realization of the present proposal is offered by putting altermagnetism on a programmable quantum simulator [24]. In optical-lattice and quantum-gas-microscope platforms, local spin preparation, single-site addressing, and repeated imaging can provide direct access to the propagation of a localized excitation in real space and time. This makes it possible to measure not only the overall spreading boundary but also the internal interference structure and its evolution as the exchange parameters are tuned. The combination of controlled J_1 - J_2 interactions and site-resolved detection therefore provides a concrete route to testing the predicted rotation of high-velocity features across the frustrated and unfrustrated regimes. More broadly, our results show that the characteristic spacetime interference generated by momentum-dependent altermagnetic magnon velocities can serve as a direct dynamical signature of symmetry-driven magnetic order, providing a bridge between momentum-space spectroscopy in solid-state materials and real-space, real-time measurements in synthetic quantum systems.

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Appendix A: The linear spin wave Hamiltonian and the Bogoliubov transformation

Starting from the spin Hamiltonian in Eq. (1) of the main text and retaining terms up to quadratic order in the Holstein–Primakoff bosons, we obtain the linear(ised) spin-wave Hamiltonian H :

$$\begin{aligned}
 H = & 2J_1 S \sum_{\langle ij \rangle} \left[a_i^\dagger a_i + b_j^\dagger b_j + a_i b_j + a_i^\dagger b_j^\dagger \right] \\
 & - 2J_2 S \sum_{\langle\langle ij \rangle\rangle} \left[a_i^\dagger a_i - a_i a_j^\dagger - a_j a_i^\dagger \right] \\
 & - 2J_2 S \sum_{\langle\langle ij \rangle\rangle} \left[b_i^\dagger b_i - b_i b_j^\dagger - b_j b_i^\dagger \right] \\
 & + N_s S^2 (J_2 - 2J_1).
 \end{aligned} \tag{A1}$$

Then, by introducing the Fourier transform on each sub-lattice, we obtain the $\mathbf{k}$ -space Bogoliubov-deGennes Hamiltonian $H_{\mathbf{k}}$. The magnon branches are then obtained by solving for the eigenmodes and eigenfrequencies of $\sigma_{\mathbf{k}}^z H_{\mathbf{k}}$ —yielding:

$$H = N_s S (S + 1) (J_2 - 2J_1) + \sum_{\mathbf{k}, \sigma} \omega_{\mathbf{k}\sigma} \left(\alpha_{\mathbf{k}\sigma}^\dagger \alpha_{\mathbf{k}\sigma} + \frac{1}{2} \right), \tag{A2}$$

where $\sigma = \pm 1$, and the corresponding magnon modes are given by:

$$\begin{pmatrix} \alpha_{\mathbf{k}, -} \\ \alpha_{-\mathbf{k}, +}^\dagger \end{pmatrix} = \begin{pmatrix} u_{\mathbf{k}} & -v_{\mathbf{k}} e^{i\varphi_{\mathbf{k}}} \\ -v_{\mathbf{k}} e^{-i\varphi_{\mathbf{k}}} & u_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ b_{-\mathbf{k}}^\dagger \end{pmatrix}, \tag{A3}$$

with $\varphi_{\mathbf{k}} = \arg(C_{\mathbf{k}})$ and the Bogoliubov coefficients $u_{\mathbf{k}}$, $v_{\mathbf{k}}$ as well as $A_{\mathbf{k}}$, $B_{\mathbf{k}}$, $C_{\mathbf{k}}$ factors defined in the main text. **Appendix B: Derivation of the excitation density for an excitation on the B sublattice**

We derive the equations governing the evolution of the system following a local excitation on the B sublattice, i.e. for $\mathbf{r}_0 \in B$. Then, for $\mathbf{r} \in B$, we have

$$\begin{aligned}
 |\phi(\mathbf{r}, t)\rangle = & \frac{1}{N_c} \sum_{\mathbf{k}} e^{-i(\mathbf{k} \cdot \mathbf{r} + \omega_{\mathbf{k}, +} t)} u_{\mathbf{k}}^2 |\emptyset\rangle \\
 & + \frac{1}{N_c} \sum_{\mathbf{k}, \mathbf{k}'} e^{-i(\mathbf{k}' \cdot \mathbf{r} - \mathbf{k} \cdot \mathbf{r}_0 + \omega_{\mathbf{k}, +} t)} \\
 & \times u_{\mathbf{k}} v_{\mathbf{k}'} |\alpha_{\mathbf{k}, +}, \alpha_{\mathbf{k}', -}\rangle.
 \end{aligned} \tag{B1}$$

For $\mathbf{r} \in A$, we instead have

$$\begin{aligned}
 |\phi(\mathbf{r}, t)\rangle = & \frac{1}{N_c} \sum_{\mathbf{k}, \mathbf{k}'} e^{-i(\mathbf{k}' \cdot \mathbf{r} - \mathbf{k} \cdot \mathbf{r}_0 + \omega_{\mathbf{k}, +} t)} \\
 & \times \sqrt{1 + \delta_{\mathbf{k}, -\mathbf{k}'}} u_{\mathbf{k}} v_{\mathbf{k}'} |\alpha_{\mathbf{k}, +}, \alpha_{-\mathbf{k}', +}\rangle.
 \end{aligned} \tag{B2}$$

The corresponding excitation density can then be written as

$$\begin{aligned}
 \rho_B(\mathbf{r}, t) = & \frac{1}{N_c^2} \left| \sum_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} + \omega_{\mathbf{k}, +} t)} u_{\mathbf{k}}^2 \right|^2 \\
 & + \frac{2}{N_c^2} \left(\sum_{\mathbf{k}} u_{\mathbf{k}}^2 \right) \left(\sum_{\mathbf{k}} v_{\mathbf{k}}^2 \right) \\
 & + \frac{2}{N_c^2} \left(\sum_{\mathbf{k}} u_{\mathbf{k}}^2 v_{\mathbf{k}}^2 \right) \\
 & + \frac{1}{N_c^2} \left| \sum_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} + \omega_{\mathbf{k}, +} t)} u_{\mathbf{k}} v_{\mathbf{k}} \right|^2.
 \end{aligned} \tag{B3}$$